

Narrative Contagion at the Sovereign Default Boundary

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Abstract

A fixed pessimistic belief has little price relevance for a safe sovereign or for one already near certain default. It matters at the boundary between them. I place this force in a sovereign bond model with narrative diffusion and debt-service feedback. Investors first become aware of a claim and then accept it after a noisy verification signal. The state therefore satisfies $0 \leq n_t \leq a_t \leq 1$, where a_t is awareness and n_t is belief. Exact one-year risk-neutral pricing maps default probability into the bond yield. The primitive belief gap is single-peaked, and a derivative condition preserves this hump in equilibrium. Numerical continuation finds 3 steady states, classifies their eigenvalues, and locates the belief seed that separates survival from default. At the boundary experiment, the threshold is 1.90 percent and the baseline two percent seed crosses it. Ten percentage points of belief raise the spread by at most 198 basis points. The two-year default wedge peaks at 2.1 percentage points. A rank-two measurement system uses media, search, analyst, and survey indicators to identify awareness and belief separately.

JEL codes: D83, D84, E71, F34, G01, H63.

Keywords: narratives, epidemiology, sovereign debt, default boundary, doom loop, text data.

1 Introduction

On July 26, 2012, the president of the European Central Bank stated that the ECB was ready to do whatever was necessary within its mandate to preserve the euro. The ECB announced the Outright Monetary Transactions program in September and never activated it. Italian and Spanish short yields fell sharply around these announcements (Altavilla et al., 2016). OMT changed redenomination insurance, conditional policy support, and the expected regime (Krishnamurthy et al., 2018). It provides a descriptive case of how a backstop can change both prices and crisis narratives.

This paper asks when a pessimistic claim can move a sovereign spread. The answer comes from the default boundary. A fixed belief distortion changes few decisions when default is remote. It also changes little when default is already nearly certain. Between these tails lies a narrow

region in which beliefs move risk-neutral default compensation, borrowing costs, and future repayment capacity.

The direct belief gap is $G(z) = \Phi(z + \Delta) - \Phi(z)$. It is the probability mass between the ordinary and pessimistic default thresholds. This gap has one peak at $z = -\Delta/2$ and vanishes in both tails. The equilibrium effect multiplies G by the spread-feedback term. Proposition 2 gives the derivative condition under which feedback preserves that peak, and the quantitative scenario verifies it on the full state space.

The contagion block separates attention from acceptance. Encounter with a believer can make an investor aware. The investor then receives a noisy verification signal and accepts the claim only when its posterior odds cross zero. Awareness and belief follow distinct continuous-time laws. This separation permits media and search measures to load mainly on awareness while investor surveys and analyst positions load on belief.

When $R_0 < 1$, a small belief seed decays. When $R_0 > 1$, it grows while fundamentals remain in the susceptible interval. Debt-service feedback then changes the fundamentals path. The full state is (θ, a, n) . Numerical continuation finds a lower no-story root at 0.0382, a positive-belief saddle at $\theta = 0.0458$, and a stable safe state at 0.1493. Along the initial-awareness curve used in the experiment, a 1.90 percent belief seed separates survival from default. The baseline seed crosses capacity zero in month 219, while the unseeded economy converges to the safe state.

The model moves prices more than default frequencies. At the hump’s peak, ten percentage points of belief raise the spread by 198 basis points. Along the seeded boundary path, the excess spread reaches 402 basis points. The same mechanism raises the two-year default probability by at most 2.1 percentage points at baseline feedback and by 1.1 to 5.3 points across the stable financing range. For a safe sovereign the addition is 0.03 points.

Policy can reduce encounters, raise belief abandonment, or cap the modelled bond yield. A one-time intervention delays a supercritical outbreak. Faster abandonment lowers the reproduction number. A binding cap removes the marginal narrative effect from the capped bond and truncates its debt-service feedback.

Flynn and Sastry (2024) build a theory, measurement system, and quantitative model of contagious firm narratives in a unique-equilibrium business cycle. They measure narratives in earnings calls and regulatory filings, then connect belief diffusion to firm decisions and aggregate fluctuations. This paper places contagious beliefs in sovereign debt. The default boundary creates a hump restriction on price relevance. Price relevance feeds back into diffusion incentives, and borrowing costs move the real default state. The empirical test therefore interacts filtered investor belief with predetermined boundary proximity. Text methods supply the observable media process (Gentzkow et al., 2019).

The empirical design combines GDELT with licensed full text, search intensity, analyst reports, and survey beliefs. Scheduled fiscal releases discipline timing. Cross-fitting and empirical Bayes shrinkage estimate boundary proximity before the narrative window. A joint bootstrap carries first-stage uncertainty into the hump regression. OMT remains a descriptive compound event.

2 A Sovereign, Its Creditors, and a Story

The threshold-default primitive delivers the direct pricing hump. Equilibrium single-peakedness adds a derivative restriction on the pricing feedback. Attention responds to decision value, while belief acceptance follows verification. The global analysis solves the full state system and continues its seed boundary.

2.1 Fundamentals, default, and pricing

Time is monthly and the pricing horizon is one year. A sovereign's fundamentals θ_t measure debt-service capacity in units of annual resources. Fundamentals mean-revert and are drained by borrowing costs:

$$\theta_{t+1} = \rho_m \theta_t + (1 - \rho_m) \bar{\theta} - \chi_m s_t + \varepsilon_{t+1}, \quad \varepsilon \sim N(0, \sigma_m^2), \quad (1)$$

where s_t is an annualized spread. Default is absorbing once $\theta_t < 0$. The monthly coefficients aggregate exactly to the annual law under a constant spread:

$$\rho_m = \rho^{1/12}, \quad \sigma_m = \sigma \sqrt{\frac{1 - \rho_m^2}{1 - \rho^2}}, \quad \chi_m = \chi \frac{1 - \rho_m}{1 - \rho}. \quad (2)$$

Thus twelve monthly innovations have variance σ^2 , and twelve monthly spread charges cumulate to χs after persistence is accounted for. At the annual horizon, the perceived default probability is

$$p(\theta, s) = \Phi(z(\theta, s)), \quad z(\theta, s) = \frac{-\rho\theta - (1 - \rho)\bar{\theta} + \chi s}{\sigma}. \quad (3)$$

The S-shape of p in θ is the paper's load-bearing object, and it comes free: any default technology in which a shock-buffed state crosses a threshold produces one.

A pessimistic narrative circulates among the sovereign's creditors. A believer prices the bond under fundamentals lower by ξ . With believing wealth share n_t , the market-clearing risk-neutral default assessment is

$$\bar{p}^Q(\theta, s, n) = (1 - n)\Phi(z) + n\Phi(z + \Delta), \quad \Delta \equiv \frac{\rho\xi}{\sigma}, \quad (4)$$

so belief shifts the standardized distance to default by Δ . A one-year zero-coupon bond with risk-free rate r , recovery $\mathcal{R}$, and risk-neutral default probability p^Q has price

$$q = e^{-r} [1 - (1 - \mathcal{R})p^Q].$$

Its default-compensation spread is

$$\mathcal{C}(p^Q) = -\log [1 - (1 - \mathcal{R})p^Q]. \quad (5)$$

If $p^Q = 1 - e^{-\lambda^Q}$, then $\mathcal{C}(p^Q) = (1 - \mathcal{R})\lambda^Q + O((\lambda^Q)^2)$. The quoted spread adds liquidity premium

LP and risk premium RP . Narrative belief moves $\mathcal{C}(\bar{p}^Q)$, while LP and RP remain fixed in the baseline. The physical simulation sets the market price of aggregate shock risk to zero, so $p^P = p^Q$, and sensitivity can estimate a separate risk-premium process. The within-period equilibrium solves

$$s = \mathcal{C}(\bar{p}^Q(\theta, s, n)) + LP + RP. \quad (6)$$

Lemma 1 (Pricing equilibrium and the doom-loop multiplier). *Suppose $0 < \mathcal{R} \leq 1$ and let $d(z, n) = (1 - n)\phi(z) + n\phi(z + \Delta)$. If*

$$\sup_{z \in \mathbb{R}, n \in [0, 1]} \mathcal{C}'((1 - n)\Phi(z) + n\Phi(z + \Delta)) \frac{\chi}{\sigma} d(z, n) < 1, \quad (7)$$

equation (6) has a unique solution for every (θ, n) . The spread rises with belief and falls with fundamentals. Small probability changes receive the multiplier

$$M(\theta, n) = \left[1 - \mathcal{C}'(\bar{p}^Q) \frac{\chi}{\sigma} d(z^*, n) \right]^{-1} \geq 1,$$

which reaches 2.0 at the reference maximum. The maximal fixed-point gain is 0.49.

The scenario sits inside the uniqueness region with maximal loop gain 0.49.

2.2 The contagion block

Let a_t denote the share aware of the narrative and let n_t denote the share that believes it. Believers form a subset of aware investors. An encounter with a believer generates attention at rate

$$\kappa(\theta, n) = \bar{m}AH_c \left(\lambda \left| \frac{\partial s^*}{\partial n} \right| \right). \quad (8)$$

After becoming attentive, investor i observes a verification signal $r_i = \ell_i + u_i$, where ℓ_i is the prior log odds in favor of the claim and u_i is noisy evidence. The investor accepts the claim when $r_i \geq 0$. Let $q = \Pr(r_i \geq 0)$. A false claim can therefore attract believers when priors or verification noise place positive mass above zero. Awareness and belief follow

$$\dot{a}_t = \kappa(\theta_t, n_t)n_t(1 - a_t) - \gamma_a a_t, \quad \dot{n}_t = q\kappa(\theta_t, n_t)n_t(1 - a_t) - \gamma_n n_t. \quad (9)$$

The reference calculation sets $q = 0.60$, $\gamma_a = 0.15$, and $\gamma_n = 0.30$ per month. The simulation advances (9) with fourth-order Runge–Kutta integration and verifies convergence across 8, 16, and 32 substeps.

Proposition 1 (Invariant cognitive state). *Suppose $0 \leq q \leq 1$, $\kappa \geq 0$, and $\gamma_n \geq \gamma_a \geq 0$. The set*

$$\mathcal{K} = \{(a, n) : 0 \leq n \leq a \leq 1\}$$

is forward invariant under (9). At zero belief, no belief appears without a seed. At full awareness, awareness falls. Along $a = n$, the vector field points into $n \leq a$.

At zero awareness and belief, the linearized belief equation grows at rate $q\kappa(\theta, 0) - \gamma_n$. For $\gamma_n > 0$, the reproduction number is

$$R_0(\theta) = \frac{q\bar{m}AH_c(\lambda v(\theta, 0))}{\gamma_n}, \quad v(\theta, n) = \frac{\partial s^*(\theta, n)}{\partial n}. \quad (10)$$

Price relevance governs attention. Verification beliefs govern acceptance. Their product determines whether a small belief seed grows.

3 The Boundary Theorems

3.1 Where stories are priced

Theorem 1 (Primitive pricing hump). *Let $G(z) = \Phi(z + \Delta) - \Phi(z)$ for $\Delta > 0$. The function G is strictly increasing on $(-\infty, -\Delta/2)$, strictly decreasing on $(-\Delta/2, \infty)$, and converges to zero in both tails. Its unique maximum is $2\Phi(\Delta/2) - 1$. Because $z^*(\theta, n)$ is strictly decreasing in θ , the direct belief gap $G(z^*)$ is single-peaked in fundamentals.*

The equilibrium price impact is

$$\frac{\partial s^*}{\partial n} = \mathcal{C}'(\bar{p}^Q) [\Phi(z^* + \Delta) - \Phi(z^*)] M(\theta, n).$$

Proposition 2 (Sufficient condition for an equilibrium hump). *Put $P(z, n) = (1 - n)\Phi(z) + n\Phi(z + \Delta)$ and $d(z, n) = P_z(z, n)$. Define*

$$Q(z, n) = \frac{\partial}{\partial z} \log \left[\frac{\mathcal{C}'(P(z, n))G(z)}{1 - \mathcal{C}'(P(z, n))(\chi/\sigma)d(z, n)} \right].$$

On an index interval $\mathcal{Z}$, suppose the pricing gain lies below one and $Q_z(z, n) < 0$. The equilibrium price impact has at most one interior peak on the corresponding fundamentals interval. It has exactly one peak if Q takes both signs. In the reference calculation, the largest value of Q_z on $z \in [-6, 6]$ and $n \in [0, 1]$ is -0.594 . The equilibrium effect is therefore single-peaked on the computed state space.

The theorem concerns the direct belief gap. The proposition states the additional restriction required for equilibrium feedback to preserve one peak. Boundedness of the multiplier alone is not sufficient.

3.2 Where stories spread

Attention rises with equilibrium price impact through equation (8). The interval result therefore uses Proposition 2.

Proposition 3 (The susceptible zone). *Suppose $\gamma_n > 0$, Proposition 2 holds, and H_c is strictly increasing over the attained values. Then $R_0(\theta)$ is single-peaked with vanishing tails. The*

susceptible set $Z = \{\theta : R_0(\theta) \geq 1\}$ is an interval. It is nonempty if and only if $q\bar{m}AH_c(\lambda v_{\max}) \geq \gamma_n$. A rise in A expands every superlevel set. A rise in γ_n contracts it. Changes in verification quality act through q .

Reducing amplification to one quarter of baseline shrinks the susceptible interval to width 0.000 in capacity units. At baseline the interval maps into spreads between 286 and 4529 basis points.

Corollary 1 (Boundary proxy). *For small Δ , the direct belief gap satisfies $G(z) = \Delta\phi(z) + O(\Delta^2)$. The lagged local sensitivity of spreads to scheduled fundamental surprises therefore supplies a proxy for boundary proximity. Recovery, risk premia, and the feedback multiplier enter the first-stage measurement model separately.*

3.3 From local drift to global paths

The state (θ_t, a_t, n_t) moves through a three-dimensional region. At zero belief, a small seed decays where $R_0 < 1$ and grows where $R_0 > 1$. Global behavior depends on the nullclines and their stability.

Proposition 4 (Instantaneous drift boundary). *Let*

$$D(\theta, n) = (1 - \rho_m)(\bar{\theta} - \theta) - \chi_m s^*(\theta, n).$$

Suppose that for each n in an interval N , the equation $D(\theta, n) = 0$ has a lower root $\hat{\theta}(n) > 0$ with $D_\theta(\hat{\theta}(n), n) > 0$. Then $\hat{\theta}$ is continuously differentiable and

$$\hat{\theta}'(n) = \frac{\chi_m s_n^*(\hat{\theta}(n), n)}{D_\theta(\hat{\theta}(n), n)} > 0.$$

For a fixed belief share, this root is the instantaneous lower drift boundary for the fundamentals equation.

The numerical continuation solves the complete three-dimensional steady-state system. It finds 3 intersections. The lower no-story state has two positive eigenvalues. The positive-belief state has awareness 53.9 percent, belief 16.2 percent, and largest eigenvalue 0.0288. It is a saddle. The safe no-story state has largest eigenvalue -0.0084 and is locally stable. The cognitive state set in Proposition 1 is forward invariant, and default is absorbing.

To trace the basin boundary, the algorithm fixes initial awareness at four times initial belief and bisects on the belief seed for each $\theta_0 \in [0.01, 0.15]$. At $\theta_0 = 0.045$, the 240-month threshold is 1.90 percent. Extending the classification horizon to 360 months changes it by 0.099 percentage points. Moving from 16 to 32 Runge–Kutta substeps changes it by 0.000 points. The baseline two percent seed lies above the boundary and defaults in month 219. The matched unseeded path converges to the safe state.

4 Policy Experiments

Proposition 5 (Policy effects within the model). *Suppose $\gamma_n > 0$. If $R_0 > 1$ is constant while belief remains near zero, removal of fraction f delays return to the original belief share by*

$$\frac{\log[1/(1-f)]}{\gamma_n(R_0 - 1)}.$$

A rise in γ_n lowers R_0 pointwise. A perfectly credible cap on the modelled bond sets $\partial s^/\partial n = 0$ wherever it binds. When $H_c(0) = 0$, new awareness through this price-relevance channel stops. Awareness and belief then decay at rates γ_a and γ_n . The cap also truncates the feedback from that bond spread to fundamentals.*

The model cap isolates the sovereign bond pricing channel. CDS, other maturities, auction quantities, capital flight, bank funding, and political choices provide additional channels. A policy backstop also changes expected recovery and the policy regime. The calculation below holds those margins fixed, while the OMT event study measures their joint movement.

If attention tracks decision value, a binding cap lowers media diffusion through this channel. A perceived-mispricing mechanism instead permits coverage to persist while the capped yield remains stable. Post-OMT text describes which pattern occurred, while the event study treats OMT as a joint change in tail insurance and the policy regime.

5 Quantitative Magnitudes

5.1 Reference calculation

Table 1 lists the reference inputs. Recovery follows the sovereign haircut evidence in Cruces and Trebesch (2013). The annual parameters map into $\sigma_m = 0.0151$ and $\chi_m = 0.0105$ through equation (2). The feedback parameter has the cash-flow decomposition

$$\chi_{it} = \text{RefinancingShare}_{it} \times \text{MarketRatePassThrough}_{it} \times \text{FiscalCapacityConversion}_{it}. \quad (11)$$

The reference value 0.12 lies inside the stable sensitivity range $[0.10, 0.20]$. Remaining maturity, the market-rate share of maturing debt, official-loan insulation, cash buffers, and fiscal response identify its factors. World Bank QPSD supplies public debt stocks and maturity fields for reporting countries (World Bank, 2026). Debt-office cash-flow schedules supply country detail.

5.2 The hump, the zone, and the outbreak

Figure 1 draws Proposition 1. Price relevance peaks at 198 basis points per ten percentage points of belief and vanishes in both tails. The susceptible zone spans $[-0.031, 0.069]$ in fundamentals and maps into spreads between 286 and 4529 basis points. Figure 2 shows how attention amplification expands this interval.

Table 1: Reference calibration.

Parameter		Value	Category	Basis
Persistence (annual)	ρ	0.90	reference input	fundamentals persistence
Innovation sd (annual)	σ	0.05	reference input	capacity volatility
Long-run capacity	$\bar{\theta}$	0.15	reference input	capacity scale
Recovery	$\mathcal{R}$	0.60	external	haircut evidence (Cruces and Trebesch, 2013)
Feedback	χ	0.12	cash-flow target	equation (11)
Story bite	ξ	0.03	belief target	$\Delta = 0.54$
Awareness decay	γ_a	0.15/mo	diffusion target	outlet and search persistence
Belief abandonment	γ_n	0.30/mo	belief target	survey and analyst persistence
Acceptance	q	0.60	verification target	belief response after exposure
Peak reproduction	R_0	4	implied target	equation (10)
Seeds	(a_0, n_0)	(0.08, 0.02)	onset target	awareness and belief indicators
Loop gain (max)		0.49	implied	uniqueness margin, Lemma 1

Notes: recovery is externally anchored. Section 6 defines the measurement system for awareness, belief acceptance, and boundary proximity.

Figure 3 reports the first 48 months of the boundary experiment. It starts at $\theta_0 = 0.045$, awareness 0.08, and belief 0.02. The initial spread is 724 basis points. Belief reaches 19 percent in the plotted window and the excess spread reaches 402 basis points. The long-horizon path crosses zero capacity in month 219, while its unseeded comparison converges to the safe state.

5.3 The crisis wedge and the policy scorecard

Figure 5 uses eight thousand common-random-number paths per starting point. The seeded minus unseeded default-probability difference reaches 2.1 percentage points at $\theta_0 = 0.03$. It is 0.03 points at $\theta_0 = 0.15$. Raising both decay rates by one half lowers the peak difference to 1.3 points. The 400 basis point cap removes the modelled belief wedge and changes unseeded fundamentals risk. Moving cap availability from month one to month thirteen raises default probability from 54.7 to 62.4 percent.

Across the stable feedback range $[0.10, 0.20]$, the peak two-year wedge runs from 1.1 to 5.3 percentage points. Peak price relevance runs from 171 to 538 basis points per ten percentage points of belief. At the upper stable value, maximal loop gain is 0.81. At $\chi = 0.30$ the gain reaches 1.22 and crosses the contraction frontier, so this value belongs to the multiple-pricing boundary rather than the stable sensitivity distribution.

6 Measurement and Empirical Designs

6.1 Text and the latent narrative state

The public core corpus is GDELT 2.0 Global Knowledge Graph and Mentions data, downloadable through BigQuery (GDELT Project, 2015). Factiva or LexisNexis provides licensed full-text validation. Google Trends measures search attention (Google, 2026). Consensus forecasts and analyst commentary measure professional beliefs. Survey probability questions supply a direct

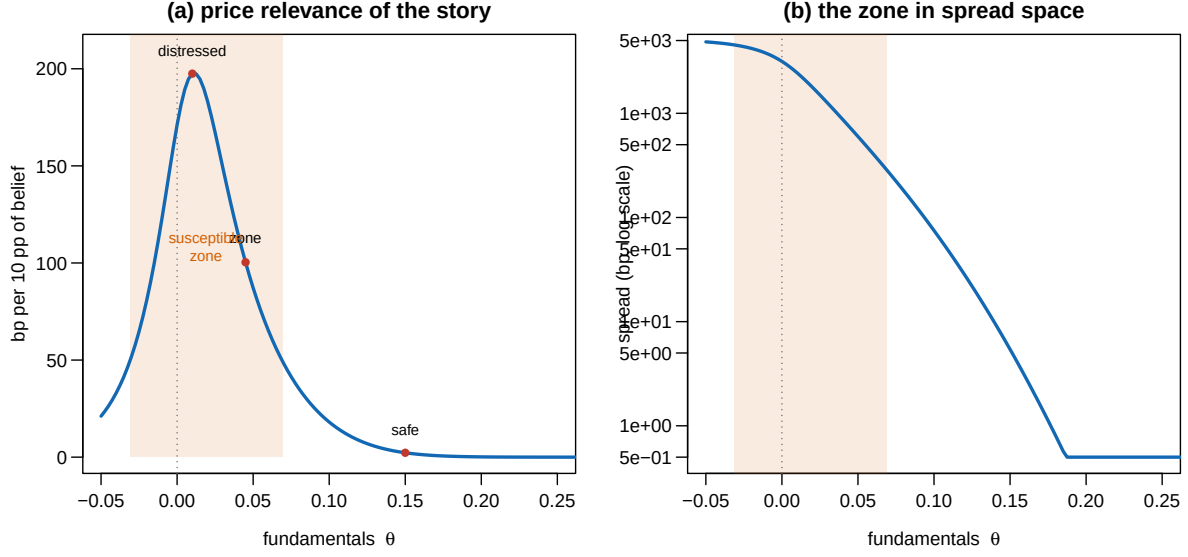


Figure 1: Equilibrium price impact and the exact one-year yield spread. Shading marks the susceptible interval.

belief indicator.

A narrative is a structured claim. It names an economic state, gives a causal explanation, implies a future sovereign outcome, and carries an action or policy implication. A high-recall dictionary screen selects candidates. A multilingual classifier assigns the structured label. Two coders label the validation sample under blinded assignment. The research design freezes the prompt, model version, label rules, and country-language error matrices before outcome analysis.

Let Z_{it}^r denote indicator r . The measurement system is

$$Z_{it}^r = c_i^r + \lambda_a^r a_{it} + \lambda_n^r n_{it} + e_{it}^r, \quad r = 1, \dots, R, \quad (12)$$

together with the transition law (9). The loading matrix $\Lambda = [\lambda_a, \lambda_n]$ must have rank two. One media loading fixes the awareness scale, and a survey probability loading fixes the belief scale. GDELT, licensed press, and search intensity load primarily on awareness. Analyst direction and survey probabilities load on belief. The state-space likelihood estimates cross-loadings and indicator-specific error. Rank and normalization restrictions assign the percentage-point scale to filtered belief.

6.2 Boundary interaction and release timing

The workhorse local projection is

$$\Delta s_{i,t+h} = \alpha_i + \tau_t + f_h(\tilde{B}_{i,t-1})\hat{n}_{it} + \Gamma_h' X_{i,t-1} + u_{i,t+h}, \quad (13)$$

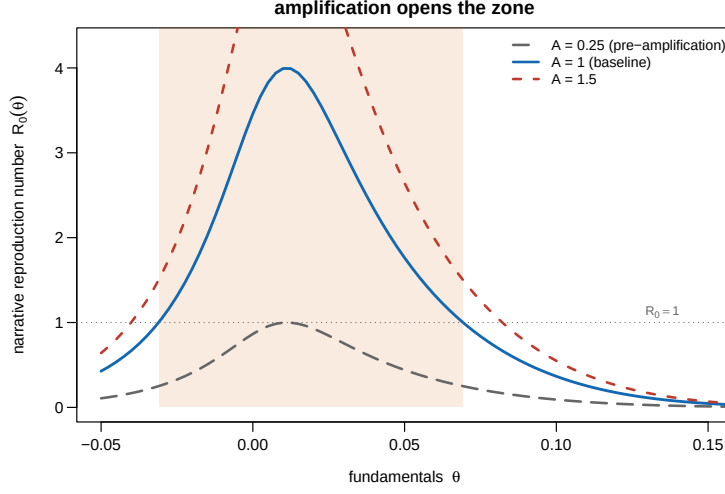


Figure 2: The narrative reproduction number at alternative amplification values. The susceptible interval is the superlevel set above one.

where the state-space model supplies filtered belief $\hat{n}_{it}$ and bins or splines estimate f_h . The theory predicts a small effect in both tails and an interior peak.

The raw boundary proxy is the lagged local sensitivity of spreads to scheduled fundamental surprises. I estimate it on a disjoint training fold, shrink the country coefficient toward a fundamentals-index prediction with empirical Bayes, and carry the posterior mean $\tilde{B}_{i,t-1}$ into the outcome fold. The fundamentals index uses debt service, reserves, maturity structure, and primary-balance forecasts. The procedure rotates folds across years and countries. A clustered bootstrap resamples release episodes, re-estimates the boundary coefficient and shrinkage rule, refilters (a, n) , and reruns (13). This joint bootstrap captures first-stage attenuation and the sampling uncertainty in the location of the hump.

For scheduled fiscal release e , estimate

$$\text{Takeup}_{ie} = a_i + \tau_e + \pi_1 \text{Surprise}_{ie} + \pi_2 \text{Surprise}_{ie} \tilde{B}_{i,e-1} + \varepsilon_{ie}, \quad (14)$$

$$\Delta s_{ie} = b_1 \text{Surprise}_{ie} + b_2 \hat{\varepsilon}_{ie} + b_3 \hat{\varepsilon}_{ie} \tilde{B}_{i,e-1} + u_{ie}. \quad (15)$$

Eurostat publishes dated calendars for government deficit and debt releases (Eurostat, 2026). These equations measure conditional timing around a known information event. Political commentary, policy responses, and market feedback can also move residual take-up, so the causal test remains equation (13) with predetermined and cross-fitted states.

6.3 Supportive variation and interpretation

Foreign media diffusion can be summarized by $N_{it}^F = \sum_{j \neq i} w_{ij,0} N_{jt}$ using predetermined language and media-overlap weights. The specification absorbs common investors, bank exposures, trade links, and regional news. The resulting estimate measures cross-border diffusion conditional on

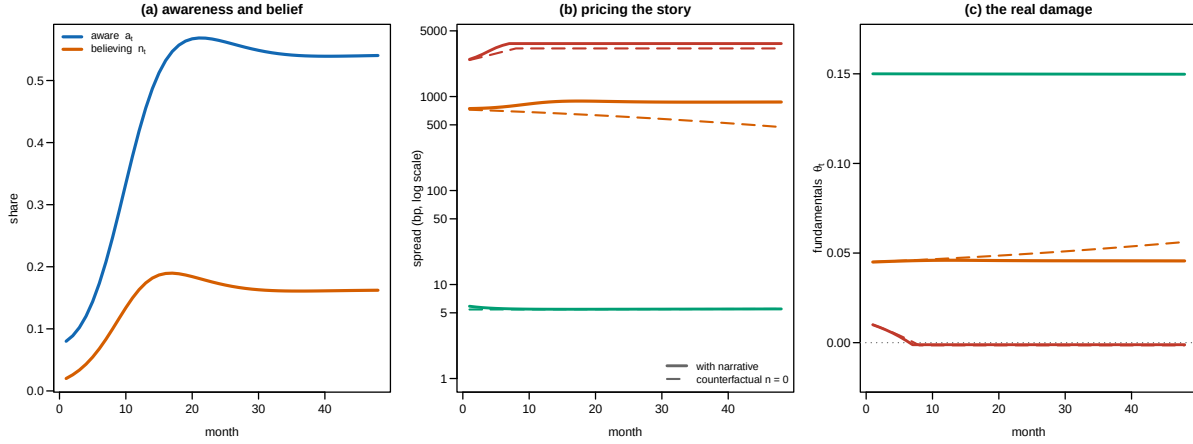


Figure 3: Deterministic boundary paths. Panel (a) separates awareness from belief. Dashed lines give the no-belief comparison in the price and fundamentals panels.

Table 2: Predictions and implementation.

Prediction	Model tude	magni- tude	Null	Estimation
Narrative pricing is hump-shaped in predetermined boundary proximity	198 vs 2.3 bp scenario		uniform effect	flexible local projections
Scheduled surprises have stronger media take-up near the boundary	Proposition 3		uniform take-up	release windows
Residual take-up has state-dependent price effects	equations (14) and (15)		no interaction	release windows
Awareness and belief load differently across indicators	rank two in equation (12)		rank one	state space
Backstop events change several channels jointly	cap scenario		pure deletion	interrupted event study
Text-price timing changes across boundary bins	dynamic fingerprint	finger- print	uniform timing	local projections

these channels.

The OMT episode enters as a descriptive interrupted event study. It compares awareness, belief indicators, diffusion speed, yields, and CDS around a backstop that jointly changed tail insurance and the expected policy regime. Lead-lag projections provide an additional timing pattern across cross-fitted boundary bins.

Table 2 separates model restrictions from reference magnitudes. The state-space rank condition and survey normalization determine the scale of belief. The joint bootstrap then carries measurement uncertainty into every market coefficient.

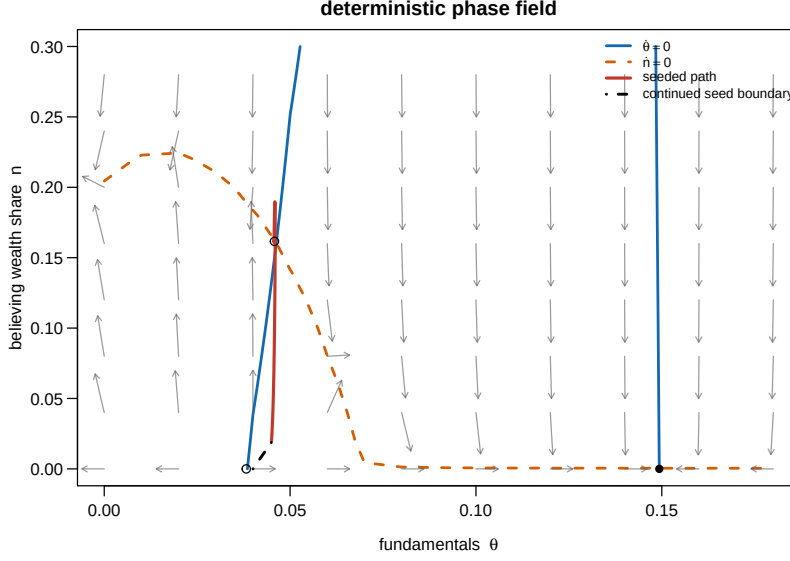


Figure 4: Projected phase field on the steady-awareness locus. The blue and orange curves are the fundamentals and belief nullclines. The red curve is the seeded path. The black dotted curve is the numerically continued seed boundary. Filled and open circles mark stable and unstable steady states.

7 Conclusion

The default boundary gives a precise restriction on narrative pricing. A fixed pessimistic shift changes risk-neutral default compensation by the probability mass between two default thresholds. This mass has one peak. Exact bond pricing and debt-service feedback preserve that peak under the derivative condition in Proposition 2.

Awareness and belief form separate states. Price relevance governs attention, while verification signals govern acceptance. Their joint dynamics remain inside $0 \leq n \leq a \leq 1$. The three-dimensional system has a stable safe state and a positive-belief saddle. Numerical continuation places the belief seed threshold at 1.90 percent in the boundary experiment.

At the reference inputs, ten percentage points of belief raise the spread by as much as 198 basis points, and seeding raises two-year default probability by 2.1 percentage points. The empirical design identifies awareness and belief from a rank-two indicator system. Cross-fitting and shrinkage estimate default-boundary proximity, while a joint bootstrap carries generated-regressor error into the hump. This combination turns a general contagion mechanism into a sovereign-specific restriction on when beliefs move prices and repayment capacity.

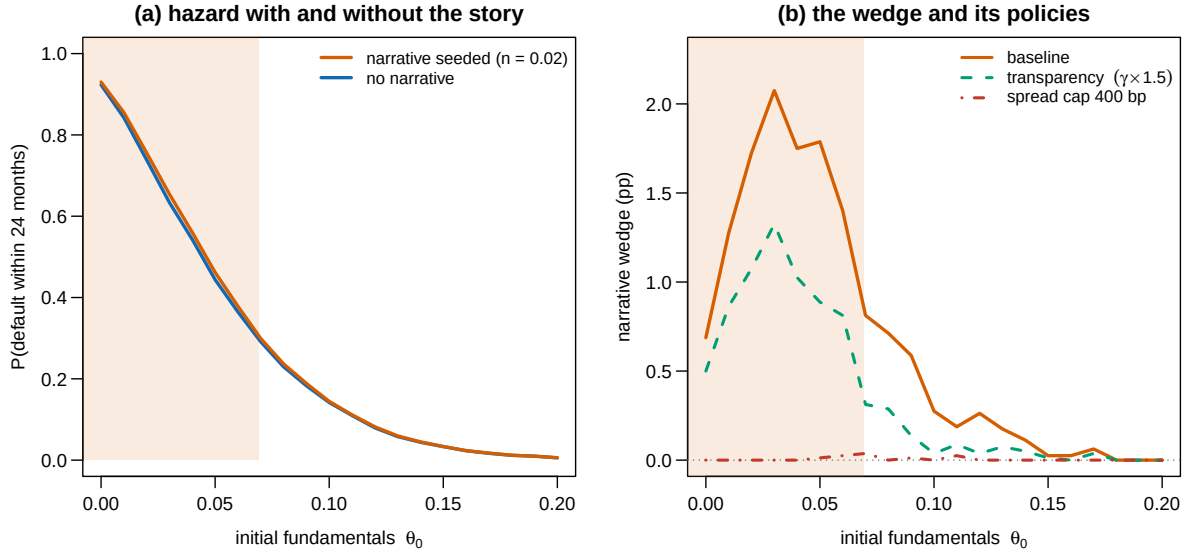


Figure 5: The narrative crisis wedge. (a) 24-month default probability by initial fundamentals, seeded against unseeded, common shocks. (b) The wedge under baseline, transparency, and a 400 bp cap (cap wedge measured against its own unseeded baseline).

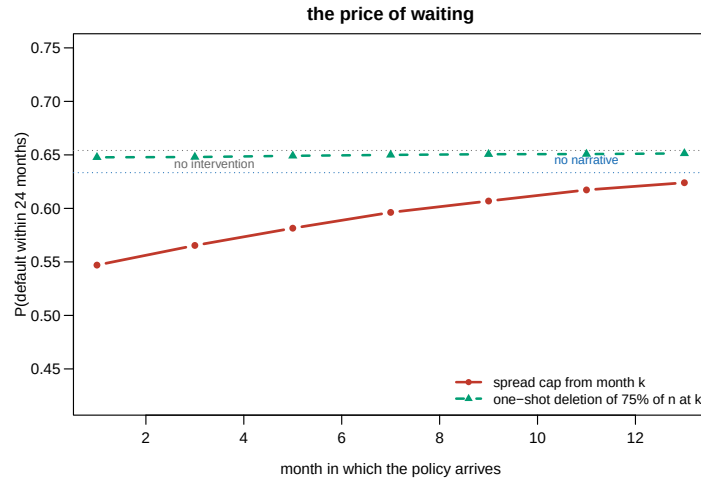


Figure 6: The price of waiting, at the wedge peak: 24-month default probability as a function of the month in which the policy arrives.

Table 3: Data and implementation.

Dataset	Coverage	Freq.	Access	Status	Use
GDEL T 2.0 GKG and global, 2015– Mentions (GDEL T Project, 2015)		15 min	public	media text	core diffu- sion panel
Factiva or LexisNexis	historical press	daily	licensed	full text	historical validation
Google Trends (Google, sovereign search terms 2026)		weekly	public	search intensity	awareness indicator
Consensus forecasts and major sovereigns commentary		monthly	licensed	analyst belief	belief indica- tor
Investor and professional available countries surveys		periodic	mixed	probability beliefs	scale nor- malization
Bond and CDS spreads	liquid sovereigns	daily	licensed	observed	licensed panel
Fiscal releases (Eurostat, EU and national 2026)		event	public	observed	surprise win- dows
Public debt (World Bank, reporting economies 2026)		quarterly	public	observed	cash-flow in- puts
ECB Data Portal (Euro- euro area pean Central Bank, 2026)		daily	public	sovereign yields	market out- comes

A Proofs

A.1 Proof of Lemma 1

Write $F(s) = \mathcal{C}(\bar{p}^Q(\theta, s, n)) + LP + RP$. Since

$$\mathcal{C}'(p) = \frac{1 - \mathcal{R}}{1 - (1 - \mathcal{R})p},$$

the fixed-point derivative is

$$F'(s) = \mathcal{C}'(\bar{p}^Q) \frac{\chi}{\sigma} d(z, n).$$

Condition (7) makes F a contraction on $[LP + RP, -\log \mathcal{R} + LP + RP]$. Banach's theorem gives a unique fixed point. Implicit differentiation gives

$$s_n^* = \mathcal{C}'(\bar{p}^Q) G(z^*) M(\theta, n) > 0, \quad s_\theta^* = -\mathcal{C}'(\bar{p}^Q) d(z^*, n) \frac{\rho}{\sigma} M(\theta, n) < 0.$$

A cap replaces F by $\min\{F, \bar{s}\}$ and preserves the contraction modulus. ■

A.2 Proof of Proposition 1

At $n = 0$, equation (9) gives $\dot{n} = 0$. At $a = 1$, $\dot{a} = -\gamma_a \leq 0$. Along the boundary $a = n$,

$$\dot{a} - \dot{n} = (1 - q)\kappa n(1 - n) + (\gamma_n - \gamma_a)n \geq 0.$$

At $a = 0$, feasibility implies $n = 0$. The vector field therefore points inward on every face of $\mathcal{K}$. Standard invariance for a locally Lipschitz vector field gives forward invariance. ■

A.3 Proof of Theorem 1

Let $G(z) = \Phi(z + \Delta) - \Phi(z) = \int_z^{z+\Delta} \phi(u) du$. Since $\phi'(u) = -u\phi(u)$,

$$G'(z) = \phi(z + \Delta) - \phi(z).$$

For $\Delta > 0$, equality $\phi(z + \Delta) = \phi(z)$ holds iff $(z + \Delta)^2 = z^2$, hence iff $z = -\Delta/2$. If $z < -\Delta/2$, then $|z + \Delta| < |z|$ and $G'(z) > 0$. If $z > -\Delta/2$, then $|z + \Delta| > |z|$ and $G'(z) < 0$. Thus G is single-peaked at $-\Delta/2$ and converges to zero in both tails. Differentiating the pricing fixed point gives

$$\frac{dz^*}{d\theta} = -\frac{\rho}{\sigma} M(\theta, n) < 0,$$

so the single-peaked profile in z maps into a single-peaked profile in fundamentals. At $z = -\Delta/2$, symmetry gives $G = 2\Phi(\Delta/2) - 1$. ■

A.4 Proof of Proposition 2

For fixed n , write the equilibrium price derivative as

$$v(z, n) = \frac{c(z, n)G(z)}{1 - c(z, n)kd(z, n)}, \quad c(z, n) = \mathcal{C}'(P(z, n)), \quad k = \chi/\sigma.$$

The contraction condition makes the denominator positive. Logarithmic differentiation gives

$$\frac{\partial \log v(z, n)}{\partial z} = Q(z, n).$$

If $Q_z < 0$, the log derivative is strictly decreasing and has at most one zero. If it takes a positive and a negative value, that zero exists and is the unique maximum. Since $z^*(\theta, n)$ is strictly decreasing in θ , the same uniqueness holds in fundamentals. For computation, use $c_z = c^2 d$ and define $B = c^2 d^2 + cd'$. Direct differentiation gives

$$Q_z = B + \frac{G''G - (G')^2}{G^2} + k \frac{B'(1 - ckd) + kB^2}{(1 - ckd)^2}, \quad B' = 2c^3 d^3 + 3c^2 dd' + cd''.$$

The grid evaluates this expression directly. ■

A.5 Proof of Proposition 3

Proposition 2 makes $v(\theta, 0)$ single-peaked with zero tails. Strict monotonicity of H_c preserves that ordering. Multiplication by $q\bar{m}A/\gamma_n$ preserves it again. A continuous single-peaked function has interval superlevel sets. The set at level one is nonempty exactly when its maximum weakly exceeds one. Increasing A raises the function pointwise and increasing γ_n lowers it pointwise, which gives the set inclusions. ■

A.6 Proof of Proposition 4

The implicit function theorem applies at the lower root because $D_\theta > 0$. Lemma 1 gives $s_n^* > 0$. Hence $D_n = -\chi_m s_n^* < 0$ and

$$\hat{\theta}'(n) = -\frac{D_n}{D_\theta} = \frac{\chi_m s_n^*}{D_\theta} > 0.$$

The numerical continuation in Section 3 supplies the corresponding global basin classification for the calibrated vector field. ■

A.7 Proof of Proposition 5

Near zero belief with constant rates, $n(t) = n(0) \exp[\gamma_n(R_0 - 1)t] + o(n(0))$. Comparing the original and reduced seeds gives the delay formula. Raising γ_n lowers $R_0 = q\kappa/\gamma_n$. Where a cap binds, the modelled spread is locally constant in n . If $H_c(0) = 0$, then $\kappa = 0$ and equation (9) reduces to $\dot{a} = -\gamma_a a$ and $\dot{n} = -\gamma_n n$. The fundamentals drift uses the capped spread. ■

B The Pricing Block: Details and the Multiplicity Frontier

Spread decomposition. Equation (5) prices one-year default compensation exactly. Writing $p^Q = 1 - e^{-\lambda^Q}$ gives

$$\mathcal{C}(p^Q) = -\log \left[\mathcal{R} + (1 - \mathcal{R})e^{-\lambda^Q} \right].$$

Its first-order term is $(1 - \mathcal{R})\lambda^Q$. Liquidity and risk premia enter additively in (6). The reference calculation sets them to zero and attributes narrative effects only to p^Q . An empirical implementation estimates both components from maturity-matched bonds and CDS.

The Calvo frontier. Condition (7) is sufficient for global contraction. The reference maximal gain is 0.49. It reaches 0.81 at $\chi = 0.20$ and 1.22 at $\chi = 0.30$. The latter crosses one and requires pricing-equilibrium selection (Calvo, 1988; Aguiar et al., 2022).

Wealth-weighted pricing. Equation (4) defines n as a wealth share. The measurement system combines quantity-weighted analyst and survey indicators with media awareness. Portfolio rebalancing can make believing wealth differ from believing headcount, which the belief loading absorbs.

C Simulation Details

Monthly parameters use the exact aggregation in equation (2). The solver computes the pricing fixed point to 10^{-12} on a 226×51 grid over $(\theta, n) \in [-0.10, 0.35] \times [0, 1]$. Bilinear interpolation supplies the within-month price and attention rate. Fourth-order Runge–Kutta integration advances awareness and belief with 16 substeps. The seed audit repeats the calculation with 8 and 32 substeps. Monte Carlo experiments use 8,000 paths of 24 months with common random numbers. Default absorbs the path at $\theta < 0$.

The steady-state routine solves the no-story fundamentals equation by complete sign-change enumeration. It solves the positive-belief system from a grid of starting values, removes duplicate roots, and computes the three-dimensional Jacobian by centered differences. Seed-boundary continuation bisection on n_0 at every initial fundamental, with $a_0 = 4n_0$. The main classification horizon is 240 months and the audit extends it to 360 months. All figures and reported values come from one JSON result file.

D Parameter Discipline and Sensitivity

Table 4 records the role of each input. Setting $\chi = 0$ removes debt-service feedback while retaining direct belief pricing. The contraction margin governs the admissible feedback range.

Table 4: Parameter discipline and sensitivity.

Parameter	Enters at	Units	Source or moment	Range exam- ined	Effect of the range
ρ, σ	eq. (3)	annual, ca- capacity	scenario, data tar- gets	joint scale	scale z and Δ jointly
$\bar{\theta} = 0.15$	eq. (1)	capacity	scenario	safe-tail an- chor	anchors the safe tail
$\mathcal{R} = 0.60$	pricing	recovery rate	Cruces and Trebesch (2013)	recovery an- chor	scales s^* and the hump
$\chi = 0.12$	eq. (11)	capacity per spread	cash-flow target	0.10 to 0.20, boundary 0.30	wedge 1.1 to 5.3 pp, boundary gain 1.22
$\xi = 0.03$	eq. (4)	capacity	scenario, belief tar- get	via Δ	hump height and width
γ_a, γ_n	eq. (9)	per month	media and belief persistence	0.15, 0.30, then $\times 1.5$	wedge 2.1 to 1.3 pp
$q = 0.60$	eqs. (9), (10)	probability	verification re- sponse	0 to 1	scales belief repro- duction
$R_0 \text{ peak} = 4$	eq. (10)	unitless	scenario, diffusion target	$A \in \{0.25, 1, 1.5\}$	zone width 0.000 to [-0.031, 0.082]
(a_0, n_0)	initial condition	latent shares	indicator onset tar- get	(0.08, 0.02)	seed threshold 1.90%
cap = 400 bp	policy	spread	OMT-scale instru- ment	month 1 to 13	hazard 54.7 to 62.4 percent
loop gain	Lemma 1	unitless	implied	0.49 to 1.22	pricing multiplicity at the boundary

Notes: recovery follows external haircut evidence. The other rows map to the data moments in Section 6.

E Data and Measurement Protocol

Sources. GDELT 2.0 supplies the public media-diffusion core (GDELT Project, 2015). Factiva or LexisNexis supplies licensed historical full text. Google Trends supplies search attention (Google, 2026). Consensus forecasts and survey probabilities measure beliefs. Eurostat supplies dated deficit and debt releases (Eurostat, 2026). World Bank QPSD and debt-office cash-flow schedules discipline the feedback block (World Bank, 2026).

Classification. A dictionary layer supplies high recall. A frozen multilingual classifier requires the structured claim defined in Section 6. A blinded, double-coded sample supplies inter-rater reliability and out-of-sample error rates by country and language.

Construction. The text index is the share of country-specific sovereign coverage carrying a classified narrative. First-appearance timestamps and the arrival of new outlets measure diffusion. Search indices add a second awareness indicator. Analyst direction and survey probabilities identify belief. Equation (12) estimates both latent states under a rank-two loading restriction (Andre et al., 2026).

F Extensions

Cross-border story networks. Let awareness and belief in country i respond to separate reach-weighted sums of foreign awareness and belief. The two-network system generates correlated spread movements among susceptible countries and supplies the model counterpart to the foreign-diffusion design (Longstaff et al., 2011).

Endogenous amplification. Attention to sovereign claims rises with volatility, making A procyclical. The susceptible zone then widens as fundamentals deteriorate. This force has the same empirical signature as a rising pricing multiplier.

Immunity and waves. Investors who rejected a claim once may resist it later. An SIRS structure with slow waning of immunity produces refractory periods and recurring waves. Conditional on boundary proximity, a repeated claim should spread more slowly on its second arrival.

Heterogeneous claims. A mixture over bite sizes ξ concentrates the priced composition toward severe claims as the boundary nears because the hump's window widens with Δ . The classification layer can test whether the severity composition rises with measured boundary proximity.

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